

Kafka Project Created

```
info : Package 'confluent.kafka' is compatible with all the specified frameworks in project 'C:\Users\Sneha\Desktop\cognizant hands-on\Week 3\Kafka-HandsOn\KafkaChatProducer\KafkaChatProducer.csproj'.
info : PackageReference for package 'Confluent.Kafka' version '2.15.0' added to file 'C:\Users\Sneha\Desktop\cognizant hands-on\Week 3\Kafka-HandsOn\KafkaChatProducer\KafkaChatProducer.csproj'.
info : Writing assets file to disk. Path: C:\Users\Sneha\Desktop\cognizant hands-on\Week 3\Kafka-HandsOn\KafkaChatProducer\obj\project.assets.json
log : Restored C:\Users\Sneha\Desktop\cognizant hands-on\Week 3\Kafka-HandsOn\KafkaChatProducer\KafkaChatProducer.csproj (in 21.58 sec).
PS C:\Users\Sneha\Desktop\cognizant hands-on\Week 3\Kafka-HandsOn\KafkaChatProducer> S
```

Zookeeper Running

```
PROBLEMS  DEBUG CONSOLE  PORTS  TERMINAL

[2026-07-05 00:26:35,536] INFO Snapshotting: 0x0 to \tmp\zookeeper\version-2\snapshot.0 (org.apache.zookeeper.server.persistence.FileTxnSnapLog)
[2026-07-05 00:26:35,543] INFO Snapshot taken in 6 ms (org.apache.zookeeper.server.ZooKeeperServer)
[2026-07-05 00:26:35,574] INFO zookeeper.request_throttler.shutdownTimeout = 10000 ms (org.apache.zookeeper.server.RequestThrottler)
[2026-07-05 00:26:35,577] INFO PrepRequestProcessor (sid:0) started, reconfigEnabled=false (org.apache.zookeeper.server.PrepRequestProcessor)
[2026-07-05 00:26:35,636] INFO Using checkIntervalMs=60000 maxPerMinute=10000 maxNeverUsedIntervalMs=0 (org.apache.zookeeper.server.ContainerManager)
[2026-07-05 00:26:35,638] INFO ZooKeeper audit is disabled. (org.apache.zookeeper.audit.ZKAuditProvider)
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```

Kafka Server Running

```
PROBLEMS  DEBUG CONSOLE  PORTS  TERMINAL

uture (kafka.server.KafkaServer)
[2026-07-05 00:29:50,562] INFO [KafkaServer id=0] End processing enable request processing future (kafka.server.KafkaServer)
[2026-07-05 00:29:50,582] INFO Kafka version: 3.9.1 (org.apache.kafka.common.utils.AppInfoParser)
[2026-07-05 00:29:50,585] INFO Kafka commitId: f745dfdcee2b9851 (org.apache.kafka.common.utils.AppInfoParser)
[2026-07-05 00:29:50,586] INFO Kafka startTimeMs: 1783191590564 (org.apache.kafka.common.utils.AppInfoParser)
[2026-07-05 00:29:50,588] INFO [KafkaServer id=0] started (kafka.server.KafkaServer)
[2026-07-05 00:29:51,168] INFO [zk-broker-0-to-controller-alter-partition-channel-manager]: Recorded new ZK controller, from now on will use node DESKTOP-9TMNS9B:9092 (id: 0 rack: null) (kafka.server.NodeToControllerRequestThread)
[2026-07-05 00:29:51,170] INFO [zk-broker-0-to-controller-forwarding-channel-manager]: Recorded new ZK controller, from now on will use node DESKTOP-9TMNS9B:9092 (id: 0 rack: null) (kafka.server.NodeToControllerRequestThread)
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```

Topic Created

```
PROBLEMS  DEBUG CONSOLE  PORTS  TERMINAL

PS C:\Users\Sneha\Desktop\cognizant hands-on> cd C:\kafka
PS C:\kafka> .\bin\windows\kafka-topics.bat --create --topic chat-topic --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
Created topic chat-topic.
PS C:\kafka> █
```

NuGet Package Installed

```
PS C:\Users\Sneha\Desktop\cognizant hands-on\Week 3> cd Kafka-HandsOn
PS C:\Users\Sneha\Desktop\cognizant hands-on\Week 3\Kafka-HandsOn> dotnet new console -n KafkaChatProducer
The template "Console App" was created successfully.

Processing post-creation actions...
Restoring C:\Users\Sneha\Desktop\cognizant hands-on\Week 3\Kafka-HandsOn\KafkaChatProducer\KafkaChatProducer.csproj:
Restore succeeded.

PS C:\Users\Sneha\Desktop\cognizant hands-on\Week 3\Kafka-HandsOn>
```

Producer Code

```
Program.cs
1  using Confluent.Kafka;
2
3  var config = new ProducerConfig
4  {
5      |   BootstrapServers = "localhost:9092"
6  };
7
8  using var producer = new ProducerBuilder<Null, string>(config).Build();
9
10 Console.WriteLine("=== Kafka Chat Producer ===");
11 Console.WriteLine("Type your message (type 'exit' to quit)");
12
13 while (true)
14 {
15     Console.Write("You : ");
16     var message = Console.ReadLine();
17
18     if (message?.ToLower() == "exit")
19         break;
20
21     await producer.ProduceAsync("chat-topic",
22         new Message<Null, string> { Value = message });
23     Console.WriteLine("Message Sent Successfully.");
24 }
25
26
```

Producer Running

```
=== Kafka Chat Producer ===
Type your message (type 'exit' to quit)
You : 
```

Consumer Code

```
Program.cs X
Program.cs
1 using Confluent.Kafka;
2
3 var config = new ConsumerConfig
4 {
5     BootstrapServers = "localhost:9092",
6     GroupId = "chat-group",
7     AutoOffsetReset = AutoOffsetReset.Earliest
8 };
9
10 using var consumer = new ConsumerBuilder<Ignore, string>(config).Build();
11
12 consumer.Subscribe("chat-topic");
13
14 Console.WriteLine("=== Kafka Chat Consumer ===");
15 Console.WriteLine("Waiting for messages...\n");
16
17 while (true)
18 {
19     var result = consumer.Consume();
20
21     Console.WriteLine($"Received : {result.Message.Value}");
22 }
```

Consumer Running

```
PROBLEMS  DEBUG CONSOLE  PORTS  TERMINAL
PS C:\Users\Sneha\Desktop\cognizant hands-on\Week 3\Kafka-HandsOn\KafkaChatConsumer> dotnet run
=== Kafka Chat Consumer ===
Waiting for messages...
█
```

Chat Demo

```
PROBLEMS  DEBUG CONSOLE  PORTS  TERMINAL
Type your message (type 'exit' to quit)
You : Hello Kafka
Message Sent Successfully.
You : Hi Sneha
Message Sent Successfully.
You : █
```

```
PROBLEMS  DEBUG CONSOLE  PORTS  TERMINAL
PS C:\Users\Sneha\Desktop\cognizant hands-on\Week 3\Kafka-HandsOn\KafkaChatConsumer> dotnet run
=== Kafka Chat Consumer ===
Waiting for messages...
Received: Hello Kafka
Received: Hi Sneha
█
```